

SkillCraft Technology Internship – Task 4

Business Insights Report (EDA on Marketing Campaign Dataset)

1. Objective

The objective of this project was to perform Exploratory Data Analysis (EDA) on a marketing campaign dataset to understand customer behavior, evaluate marketing channel performance, and recommend which marketing channels should receive a higher marketing budget based on the analysis.

2. Data Cleaning

- Loaded the marketing campaign dataset using Python and Pandas.
 - Checked for missing values and removed incomplete records.
 - Verified that there were no duplicate records.
 - Ensured all columns were in the correct format for analysis.
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3. Exploratory Data Analysis

The following analyses were performed:

- Income Distribution
- Education Level Distribution
- Marital Status Distribution
- Marketing Channel Performance (Web, Store, Catalog)
- Campaign Acceptance Analysis
- Customer Age Distribution

These visualizations helped identify customer demographics and purchasing behavior.

4. Key Findings

- Store purchases recorded the highest customer transactions.
- Web purchases also contributed significantly to overall sales.
- Catalog purchases were comparatively lower.

- Customers with higher incomes tended to spend more on products.
 - Some marketing campaigns achieved much higher acceptance rates than others.
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5. Business Recommendation

Based on the analysis, it is recommended that the company allocate a larger portion of its marketing budget to **Web** and **Store** marketing channels because they generated higher customer engagement and purchase activity. Catalog marketing should be reviewed and optimized before increasing investment. Future campaigns should focus on the customer segments that demonstrated the highest response rates to maximize return on investment (ROI).

6. Conclusion

This exploratory data analysis provided valuable insights into customer purchasing behavior and marketing effectiveness. The findings can help businesses make data-driven decisions regarding marketing budget allocation and campaign planning.